

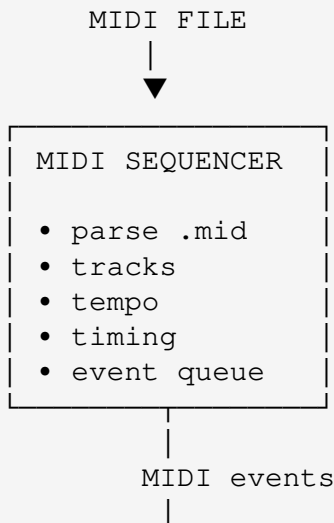
Specification

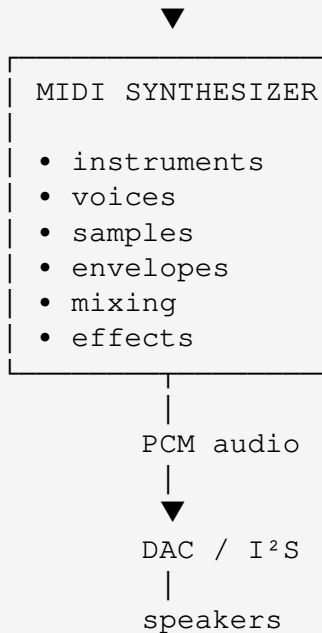
MIDI projects for generating and reading MIDI files and then play those signals as audio.

Components

- [MIDI to USB adapter](#).
- 2x [USB to MIDI](#) (both way: send and receive MIDI events)
 - Alt. <https://www.amazon.com/usb-midi/s?k=usb+to+midi>
- 2x [MIDI port](#) (DIN female 5-pin) (that can connect to breadboard)*
- Input device (keypad)*
- Optocoupler*

Block Diagram





MIDI Sequencer

- **Answers:** "What should happen and when?"
- Read MIDI file from SD card
- Produces "note data" essentially

MIDI Synthesizer

- **Answers:** "What should that event sound like?"
- Takes MIDI events and produces (synthesizes) audio:
 - Type of note
 - Note ON/OFF
 - Velocity

Types of Synthesizers

Synthesizer	Needs SF2?	How it makes sound
Oscillator synth	✗	Generates waveforms mathematically
Wavetable/ sample synth	✗ / optional	Uses samples/wavetables built into firmware or external storage
SoundFont synth	✓	Loads an .sf2 containing samples + instrument definitions

So the simplest STM32 MIDI synth could be a sine wave generator. For multiple voices ("polyphony") you can add up the sine waves.

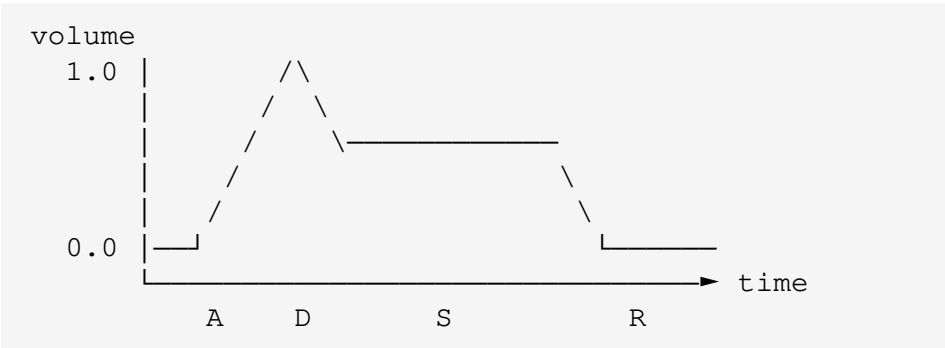
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MIDI → oscillator → envelope → mixer → DAC
```

Polyphony

Polyphony is the maximum number of individual tones or notes an electronic instrument can produce at the exact same time. When you exceed the limit, the oldest sounding notes drop out or cut off.

Envelope

If you just generate a sine wave it would start and stop abruptly, which sounds unnatural. An **envelope** applies a changing multiplier to the sine wave. The most common one is **ADSR**: Attack, Decay, Sustain, Release.

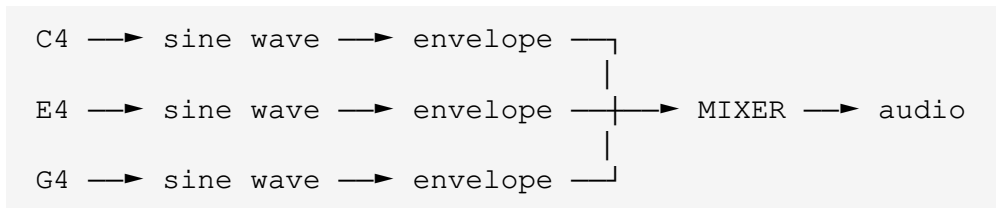


For example:

- **Attack:** how quickly the note reaches full volume
- **Decay:** how quickly it drops from full volume
- **Sustain:** volume while you hold the note
- **Release:** how quickly it fades after releasing the note

Mixer

Suppose you're playing:



The mixer simply **combines the audio signals**.